

Roshan Raj

Curriculum Vitae

Upto May 2024

Personal Details

Name: Roshan Raj
Nationality: Indian
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Research interests

Quantum Field Theory, Foundations of Quantum Theory, Relativistic Quantum Mechanics, and Quantum Dynamics.
(See: List of problems I resonate with! [Get the PDF])

Educational Details

2019-24	5 years Integrated M.Sc. (Physics), Sardar Vallabhbhai National Institute of Technology, Surat, India. CGPA: 9.33/10 (IX sem)
2016-18	Senior Secondary High Schooling, Loyola High School, CBSE, Patna, India, 77.2% [Physics, Chemistry, Mathematics, Informatics Practices, English]
2010-16	High Schooling, St. Paul's High School, ICSE, Patna, India, 91.6% , [Science, Mathematics, Computer App., Social Sci., Hindi, English]

Master's Dissertation

Title: Quantum and Classical Applications of Relativistic Relationship of Kinetic Energy and Momentum

Supervisor: Dr Vikash K Ojha, SVNIT-Surat; [Jul, 2023-May, 2024]

Brief Abstract: Reported a better relation $p^2 = (1 + \gamma)mK$, replacing the $p^2 = 2mK$ for the relativistic case and applied in quantum and classical mechanics. Predicted the contractions of Bohr radius at relativistic speeds.

Institute Mini-projects

- Title: New light on the concepts of Observable and indeterminacy in the quantum realm
Guide: Dr Vikash K Ojha, SVNIT-Surat– UG Project III, [Jul-Dec, 2022]–URL: <https://shorturl.at/6lMIS>
- Title: On the investigation of two non-neutral static bodies
Guide: Dr Sharad K Yadav, SVNIT-Surat– UG Project II, [Jan-May, 2022]– DOI: [10.48550/arXiv.2209.10641](https://doi.org/10.48550/arXiv.2209.10641)
- Title: Dynamics of Two Objects Considering the Minimum Total Potential Energy Principle
Guide: Dr. Himanshu Pandey, SVNIT-Surat– UG Project I, [Jul-Dec, 2021]–DOI: [10.31219/osf.io/ukx9t](https://doi.org/10.31219/osf.io/ukx9t)

Independent Works

- Raj, Roshan. 2023. “Obtaining the Klein-Gordon Wave Equation Without Using the Quantum Operators.” OSF Preprints. Oct 30. doi:[10.31219/osf.io/6hmyf](https://doi.org/10.31219/osf.io/6hmyf).
- Raj, Roshan. 2023. “Reviewing Observables in Classical and Quantum Mechanics.” OSF Preprints. Oct 10. doi:[10.31219/osf.io/p2ufx](https://doi.org/10.31219/osf.io/p2ufx).
- Raj, Roshan. 2023. “An Open-ended Story on Quantization.” OSF Preprints. Oct 8. doi:[10.31219/osf.io/92bvp](https://doi.org/10.31219/osf.io/92bvp).

Internship

- Title: Pramatrik Gatiki ke Chitra [*The pictures of quantum dynamics* (Hindi)]
Guide: Dr Vikash K Ojha, SVNIT-Surat – Internship I, [May-Jul, 2022]–URL: <https://shorturl.at/lvEIE>

Technical Projects

06/21-07/21	Internship, <i>The front-end development of a dummy e-commerce website</i> - Learning included using HTML5, CSS3 frameworks and training for web designing.
07/21-12/21	Project, <i>Building the student information management system using C language</i> -Learning included using C languages and its application in file handling

Achievements

09/16-present	NTS Scholarship of merit, Government of India, New Delhi.
03/24	Qualified Graduate Aptitude Test in Engineering (GATE)-2024
04/23	Qualified Graduate Aptitude Test in Engineering (GATE)-2023
02/20	Winner (Physics), InQuest 4.0, SCOSH [Student Chapter], SVNIT

Courses

University (Physics):	Quantum Field Theory I, Many-Body Physics and Relativistic Quantum Mechanics, Particle Physics, Nuclear Physics, Special Relativity, Quantum Mechanics I & II, Computational Physics(Python, Monte-Carlo), Computational Methods (MATLAB/Octave), Atomic and Molecular Physics, Electrodynamics, Statistical Mechanics, Astrophysics, Density Functional Theory, and Classical Mechanics.
University (Mathematics):	Singular perturbation theory, Special Functions, Linear Algebra, Tensors, Vector Calculus, Integral transforms, Real & Complex analysis, Computational Methods, Infinite Series, Solutions to ODEs and PDEs, Numerical methods, Interpolation, Partial Differentiation, solutions to linear equations systems etc.
Audit-courses:	Relativistic Quantum Mechanics, Classical Field Theory, Canonical Quantization, Interacting Quantum fields, Group Theory Methods in Physics [Aug-Sep'23], Introductory Lectures on Topology and Differential Geometry for Physicists [Sep-Oct'23], A Course in Math History[Aug-Sep'21], Classical Electromagnetism[Aug-Dec'20], Learning Physics through Simple Experiments[Apr-Jun'20], The advanced course on Special Theory of Relativity[Jan-May'20], The basics of Quantum Mechanics[Aug-Nov'19] and The basics of Special Theory of Relativity[Jan'18-Mar'19], TCSiON Career Edge - TCS, Mumbai [6-16th Apr'20], Life Skills for Engineers (L1), CEMCA & University of Hyderabad [Jan-Feb'20].

Soft Skills

Programme:	C, Python, MATLAB, Octave
Softwares:	L ^A T _E X, MS Excel, Mathematica, and use of GitHub.
Web Design:	Basics of HTML5, CSS3
Languages:	English (Intermediate), Hindi(Native)

Workshops & Webinars

07/21	<i>Quantum Fields, Geometry & Representation Theory</i> , ICTS-TIFR, Bangalore.
12/20	National workshop on <i>Data analysis using MS Excel</i> , BBD NITM, Lucknow.
08/20	National workshop on <i>MATLAB Tools And Applications</i> , BBD NITM, Lucknow.

Books & Talks

- 02/07/23 **MaNoVighn:** Short Poetry Collections, Kindle, ASIN: B0C9Y7X1PG
01/09/21 **KālSanGharsha:** Short Poetry Collections, Kindle, ASIN: B09FC1XRPK
13/11/21 *Unification in Mathematics*, IntERAct Seminar series, AMHD, SVNIT-Surat

Responsibilities & Positions

- 08/22-03/23 **Student Coordinator:** QUANTA Seminar Series, Department of Physics, SVNIT Surat
11/22-01/23 **Design-Infra Coordination Head:** *Manoj Memorial Night Cricket Tournament* (MMNCT) Annual cricket event, SVNIT Surat
10/21-07/22 **Joint Academic Affairs Secretary:** *Academic Affairs Council* SVNIT, Surat
03/21-07/22 **Co-convener/Co-head:** *Society for Cultivation of Science and Humanities* (SCOSH), Student Chapter, SVNIT Surat.
06/20-03/21 **Junior Graphic designer & Member:** *Society for Cultivation of Science and Humanities* (SCOSH) Student Chapter, SVNIT Surat
12/19-01/20 **Student Volunteer:** *Manoj Memorial Night Cricket Tournament* (MMNCT) Annual cricket event, SVNIT Surat
09/19-07/20 **Student Volunteer:** *Unnat Bharat Abhiyan* (UBA) SVNIT Surat

Other

Beyond being an aspiring theoretical physicist, I have always been an enthusiastic, independent seeker of truth. I will listen to mathematical beauty, ancient history, and philosophical ideas and stand up for social justice. My hobbies include Hindi Poetry, Graphic design, and writing articles.